14. Immediately after you start the stalled engine, disconnect the jump start cable from the machine that is being used as a power source. Disconnect the negative battery cable first, and then disconnect the positive battery cable.

15. Disconnect the other end of the jump start cable from the stalled machine.

16. When the engine is running and the charging system is in operation, conclude the failure analysis on the starting charging system of the stalled machine, as required.

Maintenance Section

Maintenance Access

i01978137

Access Doors and Covers

SMCS Code: 7251; 7263; 7273-572; 7273-573; 7273

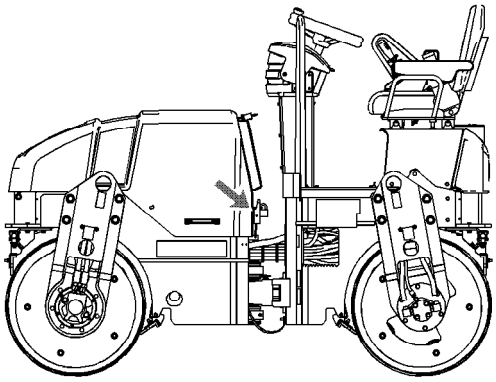


Illustration 61 g01032688

Use the handle in order to unlatch the engine hood. This will allow access to the engine and major compartments.

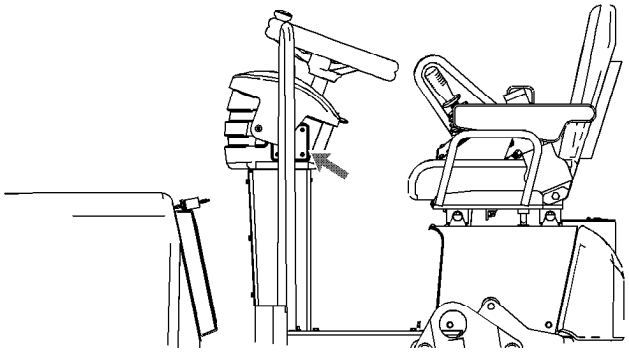


Illustration 62 g01032691

The fuses are located behind the cover plate. Remove the four screws that are holding the cover plate in place in order to gain access to the fuses.

Tire Inflation Information

i02016043

Tire Inflation with Air

SMCS Code: 4203

S/N: 2251–Up

WARNING

Use a self-attaching inflation chuck and stand behind the tread when inflating a tire.

Proper inflation equipment, and training in using the equipment, are necessary to avoid overinflation. A tire blowout or rim failure can result from improper or misused equipment.

Before inflating tire, install on the machine or put tire in restraining device.

NOTICE
Set the tire inflation equipment regulator at no more than 140 kPa (20 psi) over the recommended tire pressure.

Table 7

Tire Size	Minimum Pressure	Maximum Pressure
9.5/65-15	260 kPa (38 psi)	325 kPa (47 psi)
8.5/90-15	170 kPa (25 psi)	350 kPa (50 psi)

Note: The contact area for the tire varies 15 mm (0.6 inch) over the tire pressure range that is recommended.

i02021158

Tire Shipping Pressure

SMCS Code: 4203; 7500

S/N: 2251–Up

The shipping pressures for the machine may vary from the desired operating pressures for the machine. Adjust the tire pressures before operating the machine.

Table 8

Tire Size	Minimum Pressure	Maximum Pressure
9.5/65-15	260 kPa (38 psi)	325 kPa (47 psi)
8.5/90-15	170 kPa (25 psi)	350 kPa (50 psi)

i02610518

Tire Inflation Pressure Adjustment

SMCS Code: 4203

S/N: 2251–Up

Always obtain the proper tire inflation pressures and maintenance recommendations for the tires on your machine from your tire supplier. The tire pressure in a warm shop area 18° to 21°C (65° to 70°F) will significantly change when you move the machine into freezing temperatures. If you inflate the tire to the correct pressure in a warm shop, the tire will be underinflated in freezing temperatures. Low pressure shortens the life of a tire.

Reference: When you operate the machine in freezing temperatures, refer to Special Publication, SEBU5898, “Cold Weather Recommendations for All Caterpillar Machines” in order to adjust tire inflation pressures.

Lubricant Viscosities and Refill Capacities

i02167235

Lubricant Viscosities

SMCS Code: 1000; 7000; 7581

Table 9

Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Classification**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Engine Crankcase	**Caterpillar Multigrade DEO API CG-4 API CF-4**	SAE 0W20	−40	10	−40	50
		SAE 0W30	−40	30	−40	86
		SAE 5W30	−30	30	−22	86
		SAE 5W40	−30	40	−22	104
		SAE 10W30	−20	40	−4	104
		SAE 15W40	−15	50	5	122
Hydraulic Systems	**Caterpillar HYDO Caterpillar DEO Caterpillar TDTO Caterpillar MTO API CG-4 API CF-4 API CF Caterpillar TO-4**	SAE 0W20	−40	40	−40	104
		SAE OW30	−40	40	−40	104
		SAE 5W30	−30	40	−22	104
		SAE 5W40	−30	40	−22	104
		SAE 10W	−20	40	−4	104
		SAE 30	10	50	50	122
		SAE 10W30	−20	40	−4	104
		SAE 15W40	−15	50	5	122
		Caterpillar MTO	−25	50	5	104
		BIO HYDO-HEES(1)	−25	43	−13	110
Eccentric Weight Housing	**Synthetic ISO 220**	4C-6767 Synthetic Oil(2)	−20	50	−4	122

(1) For BIO HYDO HEES oil, tank temperatures should not exceed 100°C (212°F). Do not use BIO HYDO HEES oil in hydraulic systems with clutches or brakes.
(2) 4C-6767 Synthetic Oil is a premium synthetic gear and bearing lubricant with no viscosity improvers. This lubricant has an ISO Viscosity Grade of 220, and a minimum viscosity index of 150.

i02164915

Capacities (Refill)

SMCS Code: 1000; 6320; 7000; 7560

Table 10

REFILL CAPACITIES Approximate			
Compartment or System	**Liters**	**US Gallon**	**Imperial Gallon**

(continued)

(Table 10, contd)

Fuel Tank	48	12.7	10.6
Engine Oil	5.7	1.5	1.3
Hydraulic Tank Oil	26	6.9	5.7
Vibrator Housings	2	0.5	0.44
Water Spray Tank (Standard)	160	42.3	35.2

(continued)

(Table 10, contd)

Water Spray Tank (If Equipped)	240	63.4	52.8
Additive Spray Tank for the CB-225E	80	21.1	17.6
Coolant	6.7	1.8	1.5

i07445339

S·O·S Information

SMCS Code: 1348; 3080; 4070; 4250; 4300; 5050; 7542

S·O·S Services is a highly recommended process for Cat customers to use in order to minimize owning and operating cost. Customers provide oil samples, coolant samples, and other machine information. The dealer uses the data in order to provide the customer with recommendations for management of the equipment. In addition, S·O·S Services can help determine the cause of an existing product problem.

Refer to Special Publication, SEBU6250, “Caterpillar Machine Fluid Recommendations” for detailed information concerning S·O·S Services.

The effectiveness of S·O·S Services is dependent on timely submission of the sample to the laboratory at recommended intervals.

Refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule” for a specific sampling location and a service hour maintenance interval.

Consult your Cat dealer for complete information and assistance in establishing an S·O·S program for your equipment.

Maintenance Support

i06066971

System Pressure Release

SMCS Code: 1250-553-PX; 1300-553-PX; 1350-553-PX; 3000-553-PX; 4250-553-PX; 4300-553-PX; 5050-553-PX; 5612-553-PX; 7540-553-PX

⚠ WARNING

Personal injury or death can result from sudden machine movement.

Sudden movement of the machine can cause injury to persons on or near the machine.

To prevent injury or death, make sure that the area around the machine is clear of personnel and obstructions before operating the machine.

Coolant System

⚠ WARNING

Pressurized system: Hot coolant can cause serious burn. To open cap, stop engine, wait until radiator is cool. Then loosen cap slowly to relieve the pressure.

To relieve the pressure from the coolant system, turn off the machine. Allow the cooling system pressure cap to cool. Remove the cooling system pressure cap slowly in order to relieve pressure.

Engine Oil System

To relieve the pressure from the engine oil system, turn off the engine.

Fuel System

To relieve the pressure from the fuel system, turn off the engine.

High Pressure Fuel Lines

⚠ WARNING

Contact with high pressure fuel may cause fluid penetration and burn hazards. High pressure fuel spray may cause a fire hazard. Failure to follow these inspection, maintenance and service instructions may cause personal injury or death.

The high-pressure fuel lines are the fuel lines that are between the high-pressure fuel pump and the high-pressure fuel manifold and the fuel lines that are between the fuel manifold and cylinder head. These fuel lines are different from fuel lines on other fuel systems because of the following differences:

- The high-pressure fuel lines are constantly charged with high pressure.
- The internal pressures of the high-pressure fuel lines are higher than other types of fuel system.

Before any service or repair is performed on the engine fuel lines, perform the following tasks:

1. Turn off the engine.

2. Wait for 10 minutes.

Note: Fuel pressure can be monitored by Caterpillar Electronic Technician (ET).

Do not loosen the high-pressure fuel lines in order to purge trapped air from the fuel system.

Hydraulic System

⚠ WARNING

Personal injury can result from hydraulic oil pressure and hot oil.

Hydraulic oil pressure can remain in the hydraulic system after the engine has been stopped. Serious injury can be caused if this pressure is not released before any service is done on the hydraulic system.

Make sure all of the attachments have been lowered, oil is cool before removing any components or lines. Remove the oil filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand.

1. Shut off the engine.

2. Apply the parking brake.

3. Slowly loosen the filler cap in order to release the pressure in the hydraulic tank.

4. Tighten the filler cap.

5. The pressure in the hydraulic system has been released. Lines and components can be removed.

i07746333

Welding on Machines and Engines with Electronic Controls

SMCS Code: 1000; 7000

Do not weld on any protective structure. If it is necessary to repair a protective structure, contact your Cat dealer.

Proper welding procedures are necessary to avoid damage to the electronic controls and to the bearings. When possible, remove the component that must be welded from the machine or the engine and then weld the component. If you must weld near an electronic control on the machine or the engine, temporarily remove the electronic control to prevent heat related damage. The following steps should be followed to weld on a machine or an engine with electronic controls.

1. Turn off the engine. Place the engine start switch in the OFF position.

2. If equipped, turn the battery disconnect switch to the OFF position. If there is no battery disconnect switch, remove the negative battery cable at the battery.

NOTICE
Do NOT use electrical components (ECM or sensors) or electronic component grounding points for grounding the welder.

3. Clamp the ground cable from the welder to the component that will be welded. Place the clamp as close as possible to the weld. Make sure that the electrical path from the ground cable to the component does not go through any bearing. Use this procedure to reduce the possibility of damage to the following components:

- Bearings of the drive train
- Hydraulic components
- Electrical components
- Other components of the machine

4. Protect any wiring harnesses and components from the debris and the spatter which is created from welding.

5. Use standard welding procedures to weld the materials together.

i07746401

Maintenance Interval Schedule

SMCS Code: 1000; 7000

Ensure that all safety information, warnings and instructions are read and understood before any operation or any maintenance procedures are performed.

The user is responsible for the performance of maintenance, including all adjustments, the use of proper lubricants, fluids, filters, and the replacement of components due to normal wear and aging. Failure to adhere to proper maintenance intervals and procedures may result in diminished performance of the product and/or accelerated wear of components.

Use mileage, fuel consumption, service hours, or calendar time, WHICH EVER OCCURS FIRST, in order to determine the maintenance intervals. Products that operate in severe operating conditions may require more frequent maintenance.

Note: Before each consecutive interval is performed, all maintenance from the previous interval must be performed.

When Required

“ Battery - Recycle” 61
“ Battery or Battery Cable - Inspect/Replace”. 61
“ Circuit Breakers - Reset”. 63
“ Engine Air Filter Primary Element - Clean/ Replace” 71
“ Engine Air Filter Secondary Element - Replace” 73
“ Fuel System - Prime”. 78
“ Fuel Tank Cap and Strainer - Clean” 80
“ Fuses - Replace” 81
“ Hydraulic Oil Cooler - Clean” 82
“ Oil Filter - Inspect” 87
“ Radiator Core - Clean” 88
“ Water Spray Nozzles - Clean”. 92
“ Water Spray System - Drain” 92
“ Wheel Nuts - Tighten” 94

Every 6000 Service Hours or 3 Years

“ Cooling System Coolant Extender (ELC) - Add” 64

Every 12 000 Service Hours or 6 Years

“ Cooling System Coolant (ELC) - Change” 63

Every 10 Service Hours or Daily

“ Backup Alarm - Test” 60
“ Cooling System Coolant Level - Check”. 65
“ Drum Scrapers - Inspect/Adjust/Replace” 69
“ Engine Air Filter Service Indicator - Inspect” 74
“ Engine Oil Level - Check” 75
“ Hydraulic System Oil Level - Check ” 84
“ Indicators and Gauges - Test”. 86
“ Neutral Start Switch - Test” 86
“ Seat Belt - Inspect” 89
“ Tire Scraper - Inspect/Adjust/Replace” 91
“ Water Spray System Filter - Clean” 93

Every 50 Service Hours or Weekly

“ Tire Inflation - Check” 91

Every 100 Service Hours or 2 Weeks

“ Fuel System Water Separator - Drain” 79
“ Fuel Tank Water and Sediment - Drain” 80
“ Throttle Control - Lubricate” 91
“ Water Tank Strainer - Clean and Inspect” 94

Every 250 Service Hours

“Cooling System Coolant Sample (Level 1) - Obtain” 66

Every 250 Service Hours or 3 Months

“ Belts - Inspect/Adjust/Replace”. 61

“ Engine Oil Sample - Obtain” 76
“ Fuel System Primary Filter (Water Separator) Element - Replace” 78
“ Steering Cylinder Ends - Lubricate” 90

Initial 500 Hours (for New Systems, Refilled Systems, and Converted Systems)

“Cooling System Coolant Sample (Level 2) - Obtain” 67

Every 500 Service Hours or 3 Months

“ Engine Oil and Filter - Change”. 77

Every 500 Service Hours or 6 Months

“ Articulating and Oscillating Bearings - Lubricate” 60
“ Braking System - Test” 62
“ Hydraulic System Oil Filter - Replace” 83
“ Hydraulic System Oil Sample - Obtain” 85
“Isolation Mounts - Inspect”. 86

Every 1000 Service Hours or 1 Year

“ Battery - Clean/Check” 60
“ Cooling System Pressure Cap - Clean/ Replace” 67
“ Engine Cylinder Head Bolts - Check”. 75
“ Engine Mounts - Inspect” 75
“ Engine Valve Lash - Check” 78
“ Hydraulic System Oil - Change” 82
“ Hydraulic Tank Breather - Replace” 85
“ Hydraulic Tank Strainer - Clean”. 85
“ Propel Control Tension - Adjust”. 87
“ Rollover Protective Structure (ROPS) - Inspect”. 89

Every 1000 Service Hours or 2 Years

“ Drum Bearings - Inspect” 69

Every 2000 Service Hours

“ Engine Crankcase Breather - Clean/Replace”. . . 74

Every Year

“Cooling System Coolant Sample (Level 2) - Obtain” 67

Every 3000 Service Hours

“ Eccentric Weight Housing Oil - Change” 70

Every 3000 Service Hours or 2 Years

“ Cooling System Water Temperature Regulator - Replace” 68

Every 3 Years

“ Seat Belt - Replace”. 90

i01977833

Articulating and Oscillating Bearings - Lubricate

SMCS Code: 7051-086-BD; 7057-086-BD; 7113-086-BD

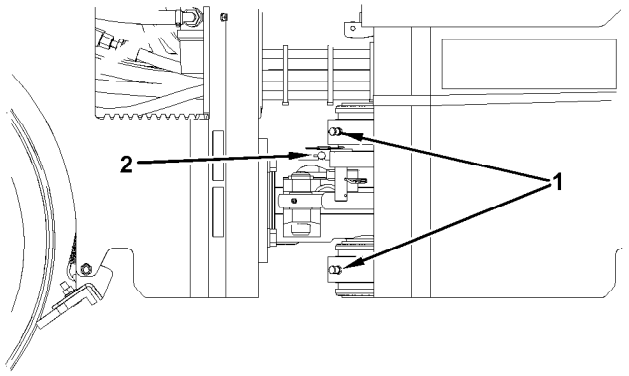


Illustration 63 g01032832

(1) Articulating Bearings
(2) Oscillating Bearing

The hitch is located in the center pivot area.

1. Clean all fittings before servicing.

2. Clean all caps before servicing.

3. Lubricate the fittings for the articulation bearing.

Note: Use a small amount of grease. Excess grease can cause seal damage.

4. Install all caps after servicing.

i01977834

Backup Alarm - Test

SMCS Code: 7406-081

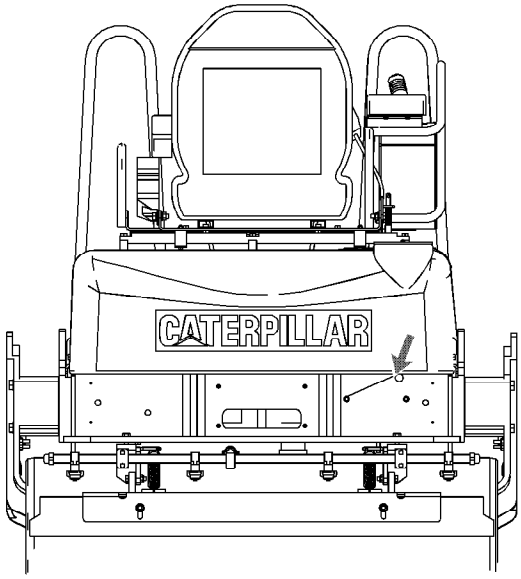


Illustration 64 g01032985

The back up alarm is located at the rear of the machine.

1. Engage the parking brake.

2. Start the engine.

3. Move the propel control lever to the REVERSE position. The backup alarm should sound immediately. The backup alarm will continue to sound until the propel control lever is moved to the NEUTRAL position or to the FORWARD position.

4. If the back up alarm does not sound, make the necessary repairs.

i01930685

Battery - Clean/Check

SMCS Code: 1401-070; 1401-535; 1402-070; 1402-535

1. Clean the top of the batteries with a clean cloth.

2. Clean the battery terminals. Coat the battery terminals with petroleum jelly, if necessary.

3. Check the battery cables. Tighten any loose connections.

i07746330

Battery - Recycle

SMCS Code: 1401-561

Always recycle a battery. Never discard a battery.

Always return used batteries to one of the following locations:

- A battery supplier
- An authorized battery collection facility
- Recycling facility

i02012676

Battery or Battery Cable - Inspect/Replace

SMCS Code: 1401-510; 1401-040; 1402-040; 1402-510

1. Turn the engine start switch to the OFF position. Turn all switches to the OFF position.

2. Open the battery compartment.

3. Disconnect the negative battery cable at the battery.

4. Disconnect the positive battery cable from the battery.

5. Remove the cable from the starter motor.

6. Perform the necessary repairs. Replace the cables or the battery, as needed.

7. Reverse the above steps in order to reconnect the battery.

8. Install the battery compartment covers.

i02370377

Belts - Inspect/Adjust/Replace

SMCS Code: 1357-510; 1357-025; 1357-040

Your engine is equipped with a belt that operates the fan, the alternator, and the water pump. Your engine is also equipped with a belt that operates the vibratory pump. For maximum engine performance and maximum utilization of your engine, inspect the belts for wear and for cracking. Check the tension of the belts. Adjust the tension of the belts in order to minimize belt slippage. Belt slippage will decrease the belt life. Belt slippage will also cause poor performance.

If new belts are installed, recheck the belt adjustment after 30 minutes of operation. If two belts or more are required for an application, replace the belts in belt sets. If only one belt of a matched set is replaced, the new belt will carry more load. This is due to the fact that the older belts are stretched. The additional load on the new belt could cause the new belt to break.

Alternator Belt and Water Pump Belt

1. Open the engine compartment.

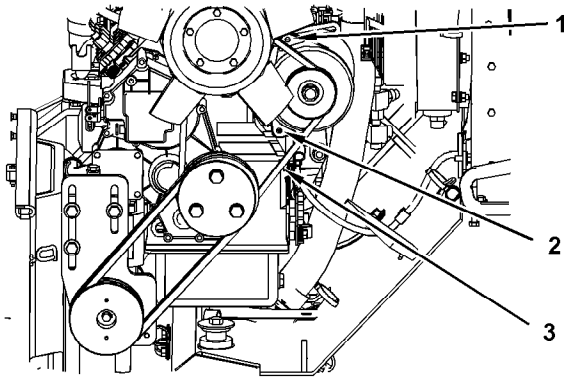


Illustration 65 g01033074

2. To check the belt tension, apply 110 N (25 lb) of force midway between the pulleys. Correctly adjusted belts will deflect 13 to 19 mm (1/2 to 3/4 inch).

3. In order to adjust alternator belt (3), loosen mounting bolts (1) and (2).

4. To achieve the correct adjustment, move the alternator inward or move the alternator outward, as required.

5. Tighten mounting bolts (1) and (2).

Note: The alternator shaft nut must be tightened to a torque of 50 ± 5 N·m (37 ± 4 lb ft).

6. If new belts are installed, check the belt adjustment again after 30 minutes of engine operation at the rated speed.

7. Close the engine compartment.

Vibratory Pump Belt

1. Open the engine compartment.

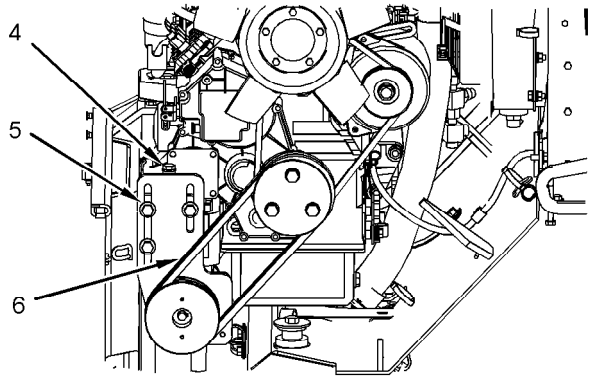


Illustration 66 g01045327

2. In order to check the belt tension, apply 40 N (9 lb) of force to the belt. The force must be applied midway between the pulleys. The belts will deflect 5 to 6 mm (.2 to .24 inch) if the belts are adjusted correctly.

3. In order to adjust vibratory pump belts, loosen three mounting bolts (5).

4. In order to tighten the belts, turn adjusting bolt (4) clockwise. In order to loosen the belts, turn adjusting bolt (4) counterclockwise.

5. Tighten three mounting bolts (5).

6. Close the engine compartment.

i02165934

Braking System - Test

SMCS Code: 4250-081; 4267-081

Note: If the machine configuration changes, the parking brakes need to be tested.

Check the area around the machine. Make sure that the machine is clear of personnel and clear of obstacles.

Put the steering frame lock in the UNLOCKED position.

Fasten the seat belt before checking the parking brake.

The following tests are used to determine if the parking brake is functional on a specified grade or a specified slope. These tests are not intended to measure the maximum brake holding effort. Read all of the steps before you perform the following procedure.

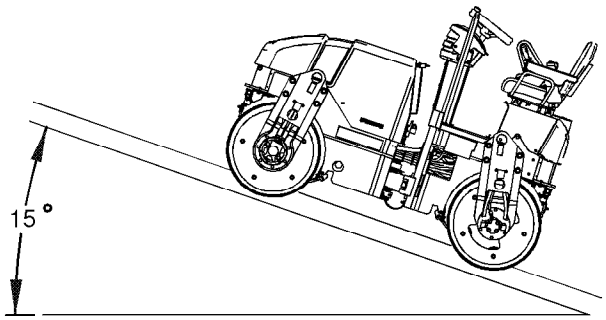


Illustration 67 g01039944

Position the machine on the incline of the slope, but near the base of the slope in order to check the parking brake. The test position should be 26 percent or a 15 degree slope.

1. Start the engine. Refer to the Operation and Maintenance Manual, “Engine Starting” for information on starting the engine.

2. Move the machine into the test position.

3. Place the throttle control into the LOW IDLE position.

4. Engage the parking brake.

The machine should not move under the following conditions.

- The engine is at low idle.
- The parking brake is applied.
- The machine is positioned on the specified slope.

⚠ WARNING

Personal injury can result if the machine moves while testing.

If the machine begins to move, release the parking brake and use the propel lever in order to move the machine to a level surface.

5. Park the machine on a level surface.

6. Stop the engine.

NOTICE
If the machine moved during the brake test, consult your Caterpillar dealer.

The dealer must inspect the brake system and make any necessary repairs before the machine is returned to operation.

i02844450

Circuit Breakers - Reset

SMCS Code: 1420-529

The circuit breaker is located in the engine compartment above the battery.

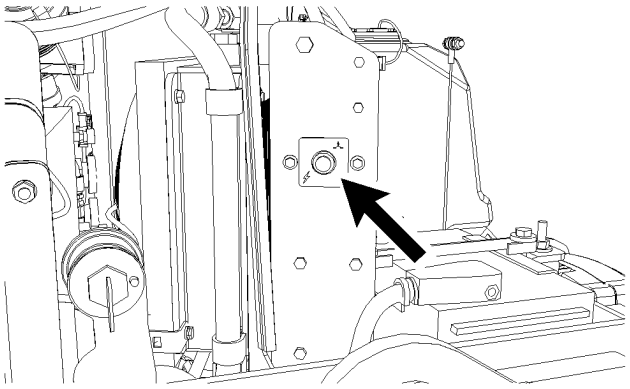


Illustration 68 g01418362

Circuit Breaker/Reset – Push in the button in order to reset the circuit breaker. If the electrical system is functioning properly, the button will remain depressed. If the button does not remain depressed, check the appropriate electrical circuit. Repair the electrical circuit, if necessary.

i01977846

Cooling System Coolant (ELC) - Change

SMCS Code: 1395-044-NL

⚠ WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, “Caterpillar Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Do not change the coolant until you read and understand the material in the Cooling System Specifications section.

NOTICE
Mixing Extended Life Coolant (ELC) with other products reduces the effectiveness of the coolant and shortens coolant life. Use only Caterpillar products or commercial products that have passed the Caterpillar EC-1 specifications for premixed or concentrate coolants. Use only Caterpillar Extender with Caterpillar ELC. Failure to follow these recommendations could result in the damage to cooling systems components.

If ELC cooling system contamination occurs, refer to Operation and Maintenance, “Extended Life Coolant (ELC)” under the topic ELC Cooling System Contamination.

Drain the coolant whenever the coolant is dirty. Drain the coolant when foam is observed.

1. Stop the engine. Allow the cooling system to cool completely.
2. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

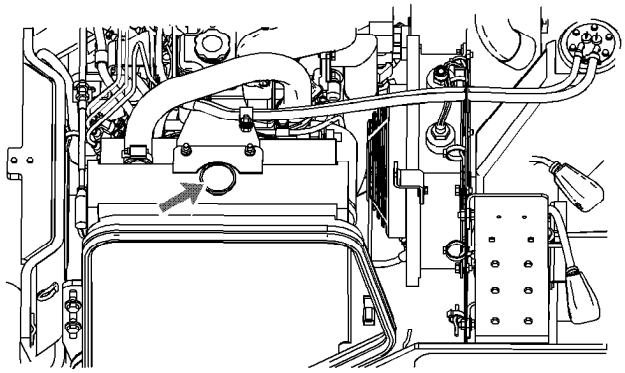


Illustration 69 g01033218

3. Slowly loosen the cooling system pressure cap in order to relieve system pressure. Remove the cooling system pressure cap.

4. Remove the lower hose that is connected to the radiator. Allow the coolant to drain into a suitable container.

5. Replace the lower drain hose. Fill the cooling system with clean water and with a 6 to 10% concentration of cooling system cleaner.

6. Install the cooling system pressure cap.

7. Close the engine compartment.

8. Start the engine. Run the engine for 90 minutes.

9. Stop the engine. Allow the cooling system to completely cool.

10. Open the engine compartment.

11. Remove the cooling system pressure cap.

12. Remove the lower drain hose from the radiator. Drain the cleaning solution.

13. Flush the cooling system with water until the draining water is transparent.

14. Replace the lower drain hose.

15. Add the recommended amount of extender to the coolant system. Refer to Operation and Maintenance Manual, “Capacities (Refill)” for the proper amount.

Note: If you are using Caterpillar Long Life Coolant that contains some additive, do not add any supplemental coolant additive at this time. Also, do not change the coolant conditioner element if you are using Caterpillar Long Life Coolant that contains some additive.

16. Start the engine and run the engine. Leave the cap off until the thermostat opens and the coolant level stabilizes.

17. Maintain the coolant level to 1 cm of the bottom of the fill pipe.

18. Inspect the gasket on the cooling system pressure cap. Replace the cooling system pressure cap if the gasket is damaged.

19. Install the cooling system pressure cap.

20. Close the engine compartment.

i01977855

Cooling System Coolant Extender (ELC) - Add

SMCS Code: 1352-544-NL

⚠ WARNING

At operating temperature, the engine coolant is hot and under pressure.

Steam can cause personal injury.

Check the coolant level only after the engine has been stopped and the fill cap is cool enough to touch with your bare hand.

Remove the fill cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Avoid contact with the skin and eyes to prevent personal injury.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, “Caterpillar Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

When a Caterpillar Extended Life Coolant (ELC) is used, an Extender must be added to the cooling system. For additional information about the addition of Extender, see the Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” or consult your Caterpillar dealer.

Use a 8T-5296 Coolant Test Kit to check the concentration of the coolant.

NOTICE
Mixing ELC with other products that do not meet Caterpillar EC-1 specifications reduces the effectiveness of the coolant and shortens coolant service life.

Use only Caterpillar products or commercial products that have passed the Caterpillar EC-1 specification for pre-mixed or concentrate coolants. Use only Caterpillar Extender with Caterpillar ELC.

Failure to follow these recommendations can result in shortened cooling system component life.

1. Stop the engine. Allow the cooling system to completely cool.

2. Open the engine compartment.

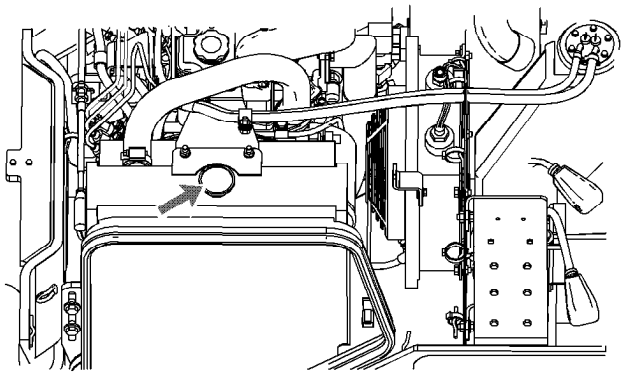


Illustration 70 g01033218

3. Slowly loosen the cooling system pressure cap in order to relieve system pressure. Remove the cooling system pressure cap.

4. If necessary, drain enough coolant from the radiator in order to allow the addition of the Extender.

5. Add the recommended amount of extender to the coolant system. Refer to the Special Publication, SEBU6250, “Extended Life Coolant (ELC)” for the proper amount.

6. Maintain the coolant level to 1 cm of the bottom of the fill pipe.

7. Inspect the gasket on the cooling system pressure cap. Replace the cooling system pressure cap if the gasket is damaged.

8. Install the cooling system pressure cap.

9. Close the engine compartment.

For additional information on the addition of extender, see Special Publication, SEBU6250, “Coolant Recommendations” or consult your Caterpillar dealer.

i01977858

Cooling System Coolant Level - Check

SMCS Code: 1350-535-FLV

⚠ WARNING

At operating temperature, the engine coolant is hot and under pressure.

Steam can cause personal injury.

Check the coolant level only after the engine has been stopped and the fill cap is cool enough to touch with your bare hand.

Remove the fill cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Avoid contact with the skin and eyes to prevent personal injury.

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

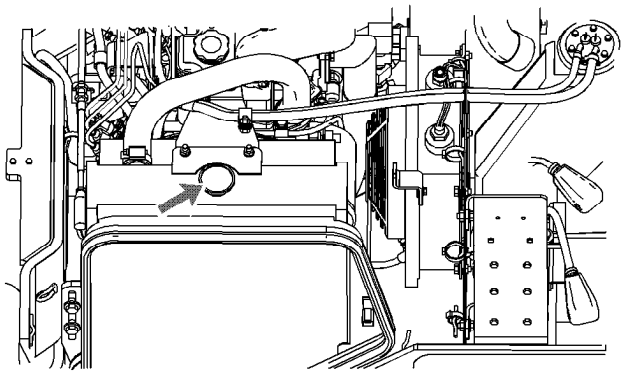


Illustration 71 g01033218

2. Loosen the radiator filler cap slowly in order to relieve pressure. After pressure is relieved, remove the radiator cap.

3. Maintain the coolant level to 1 cm of the bottom of the fill pipe.

4. Inspect the radiator filler cap and the radiator cap seal for damage. Clean the cap with a clean cloth or replace the cap.

5. Install the cap.

6. Close the engine compartment.

i02425978

Cooling System Coolant Sample (Level 1) - Obtain

SMCS Code: 1350-008; 1395-008; 1395-554; 7542; 7542-008

Note: It is not necessary to obtain a Coolant Sample (Level 1) if the cooling system is filled with Cat ELC (Extended Life Coolant). Cooling systems that are filled with Cat ELC should have a Coolant Sample (Level 2) that is obtained at the recommended interval that is stated in the Maintenance Interval Schedule.

Note: Obtain a Coolant Sample (Level 1) if the cooling system is filled with any other coolant instead of Cat ELC. This includes the following types of coolants.

- Commercial long life coolants that meet the Caterpillar Engine Coolant Specification -1 (Caterpillar EC-1)
- Cat Diesel Engine Antifreeze/Coolant (DEAC)
- Commercial heavy-duty antifreeze/coolant solution

NOTICE
Always use a designated pump for oil sampling, and use a separate designated pump for coolant sampling. Using the same pump for both types of samples may contaminate the samples that are being drawn. This contaminate may cause a false analysis and an incorrect interpretation that could lead to concerns by both dealers and customers.

Note: Level 1 results may indicate a need for Level 2 Analysis.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, “Caterpillar Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

Obtain the sample of the coolant as close as possible to the recommended sampling interval. The recommended sampling interval for Level 1 Coolant Analysis is every 250 service hours. In order to receive the full effect of S·O·S analysis, you must establish a consistent trend of data. In order to establish a pertinent history of data, perform consistent samplings that are evenly spaced. Supplies for collecting samples can be obtained from your Caterpillar dealer.

Use the following guidelines for proper sampling of the coolant:

- Complete the information on the label for the sampling bottle before you begin to take the samples.
- Keep the unused sampling bottles stored in plastic bags.
- Keep the lids on empty sampling bottles until you are ready to collect the sample.
- Place the sample in the mailing tube immediately after obtaining the sample in order to avoid contamination.
- Never collect samples from expansion bottles.
- Never collect samples from the drain for a system.

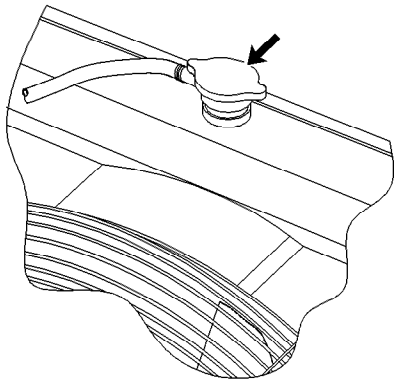


Illustration 72 g00544510

⚠ WARNING

Pressurized System: Hot coolant can cause serious burns. To open the cooling system filler cap, stop the engine and wait until the cooling system components are cool. Loosen the cooling system pressure cap slowly in order to relieve the pressure.

1. The machine needs to be operated in order to circulate the coolant. Collect the sample after a normal workday. Collect the samples from one to two hours after the engine has been shut off.
2. Start the engine momentarily in order to circulate the coolant again.
3. Shut off the engine.
4. Carefully remove the radiator cap.
5. Use a vacuum pump and draw the sample. Do not allow dirt or other contaminants to enter the sampling bottle. Fill the sampling bottle three-fourths from the top. Do not fill the bottle completely.
6. Place the sampling bottle with the completed label into the mailing tube.
7. Install the radiator cap.

i07349178

Cooling System Coolant Sample (Level 2) - Obtain

SMCS Code: 1350-008; 1395-554; 1395-008; 7542-008; 7542

Reference: Refer to Operation and Maintenance Manual, “Cooling System Coolant Sample (Level 1) - Obtain” for the guidelines for proper sampling of the coolant.

Obtain the sample of the coolant as close as possible to the recommended sampling interval. Supplies for collecting samples can be obtained from your Cat dealer.

Submit the sample for Level 2 analysis.

Reference: For additional information about coolant analysis, refer to Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” or consult your Cat dealer.

i01977864

Cooling System Pressure Cap - Clean/Replace

SMCS Code: 1382-510; 1382-070

⚠ WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

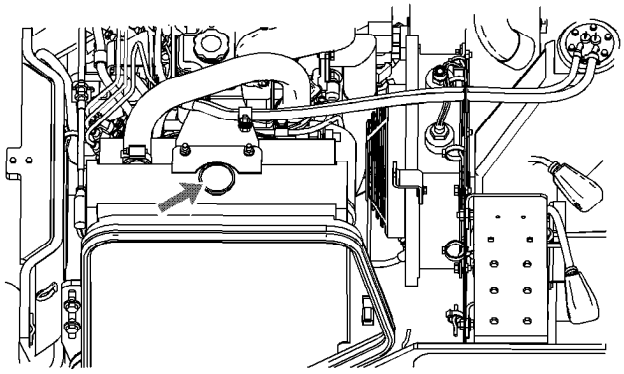


Illustration 73 g01033218

2. Remove the cooling system pressure cap slowly in order to relieve pressure.
3. Inspect the cooling system pressure cap for foreign material, for deposits, and for damage. Clean the cooling system pressure cap with a clean cloth. If the cooling system pressure cap is damaged, replace the cooling system pressure cap.
4. Install the cooling system pressure cap.

5. Close the engine compartment.

i01977885

Cooling System Water Temperature Regulator - Replace

SMCS Code: 1355-510; 1393-010

⚠ WARNING

At operating temperature, the engine coolant is hot and under pressure.

Steam can cause personal injury.

Check the coolant level only after the engine has been stopped and the fill cap is cool enough to touch with your bare hand.

Remove the fill cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Avoid contact with the skin and eyes to prevent personal injury.

Replace the water temperature regulator on a regular basis in order to reduce the chance of unscheduled downtime and of problems with the cooling system.

The water temperature regulator should be replaced after the cooling system has been cleaned. Replace the water temperature regulator while the cooling system is completely drained. Replace the water temperature regulator while the cooling system coolant is drained to a level below the water temperature regulator housing.

NOTICE
Failure to replace the engine's water temperature regulator on a regularly scheduled basis could cause severe engine damage.

Note: If you are only replacing the water temperature regulator, drain the cooling system coolant to a level that is below the water temperature regulator housing.

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

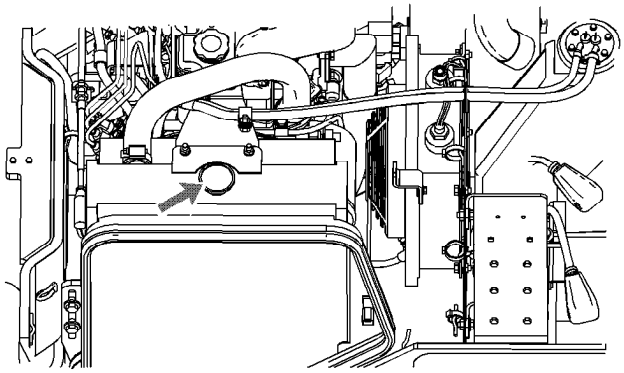


Illustration 74 g01033218

2. Remove the cooling system pressure cap in order to relieve the pressure in the cooling system.

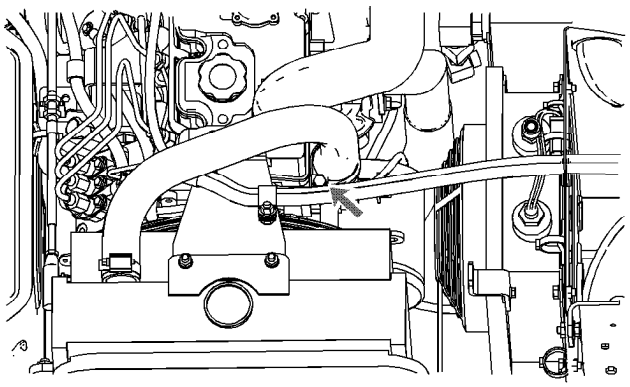


Illustration 75 g01033277

3. Remove the housing for the water temperature regulator.

4. Remove the gasket and remove the water temperature regulator.

NOTICE
Former water temperature regulators may be used, if they meet test specifications and are not damaged or have excessive buildup or deposits.

NOTICE
Since Caterpillar engines incorporate a shunt design cooling system, it is mandatory to always operate the engine with a water temperature regulator.

Depending on load, failure to operate with a water temperature regulator could result in either an overheating or an overcooling condition.

NOTICE
If the water temperature regulator is installed incorrectly, it will cause the engine to overheat.

5. Install a new water temperature regulator and install a new gasket.

6. Install the housing for the water temperature regulator.

7. Add the cooling system coolant. Maintain the level of the coolant to 1 cm from the bottom of the fill pipe.

8. Inspect cooling system pressure cap and the gasket for damage. Replace the pressure cap if the pressure cap or the gasket are damaged.

9. Install the cooling system pressure cap.

10. Close the engine compartment.

i03505504

Drum Bearings - Inspect

SMCS Code: 6605-040-BD; 7551-040-BD

The drum bearings should be inspected periodically as part of a preventive maintenance schedule. Consult your Caterpillar dealer for information concerning the inspection of the drum bearings.

For additional information, refer to Disassembly and Assembly, KENR3674, “Drum and Drum Support - Remove”.

i01977897

Drum Scrapers - Inspect/ Adjust/Replace

SMCS Code: 6607-040; 6607-510; 6607-025

Inspect Scrapers

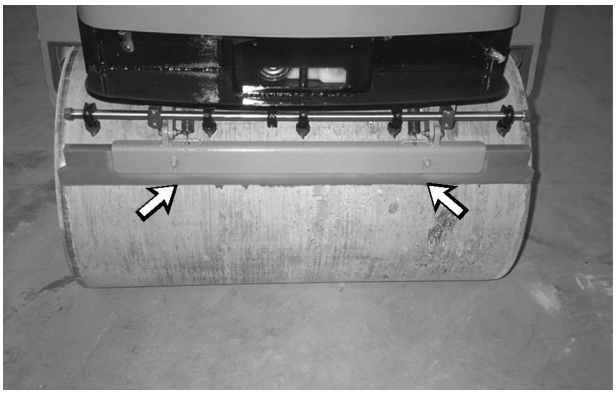


Illustration 76 g00640988



Illustration 77 g00641016

1. Remove dirt and debris from scrapers.

2. The drum scraper should be adjusted vertical to the ground. The drum scraper should be adjusted in order to touch the width of the drum. The inner drum scrapers are not adjustable in the same way as the outer drum scrapers. The inner drum scrapers can be adjusted with the adjustable spring mounts. The adjustable spring mounts are located inside the front and rear frames. The tension on the spring mounts should be adjusted in order to prevent bouncing of the tire or the drum scraper when the vibration system is active. The tire scrapers should be adjusted slightly above the surface of the tire. The scraper should not touch the tire.

Replace Scrapers

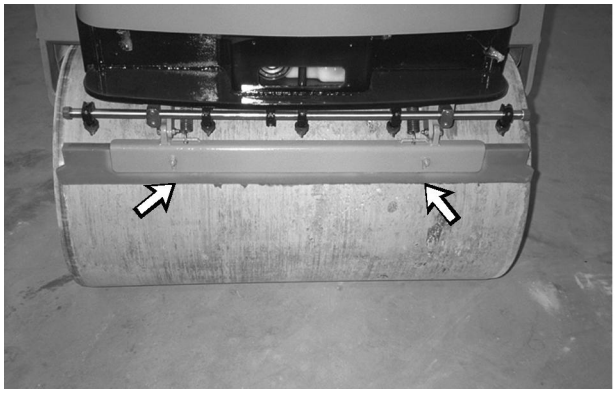


Illustration 78 g00641022

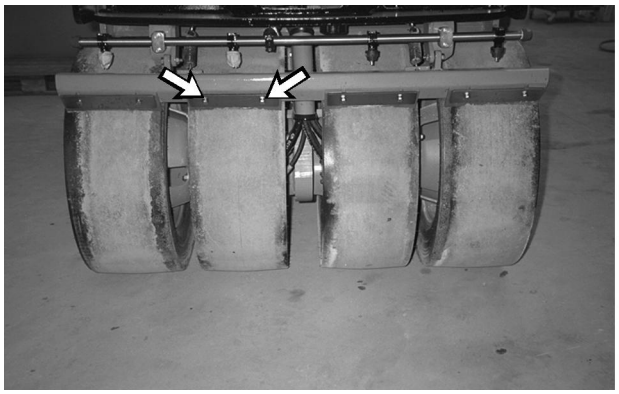


Illustration 79 g00641029

1. Loosen the bolts and remove the damaged scraper.

2. Install the new scraper and tighten the bolts.

3. Adjust the scraper in order to touch the drum.

4. Adjust the scraper slightly above the tires.

i03829386

Eccentric Weight Housing Oil - Change

SMCS Code: 6606-044-OC

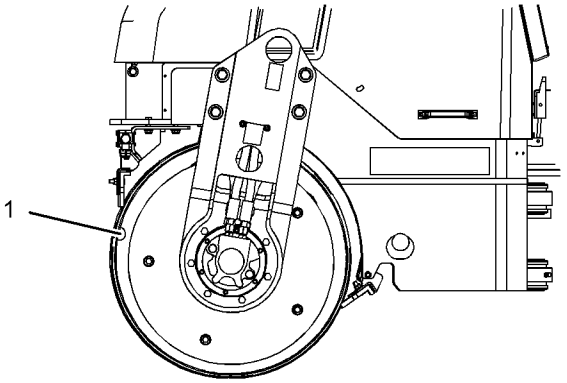


Illustration 80 g01969953

1. Orient the drum so that water drain hole (1) is at the bottom. This will properly orient the oil drain hole.

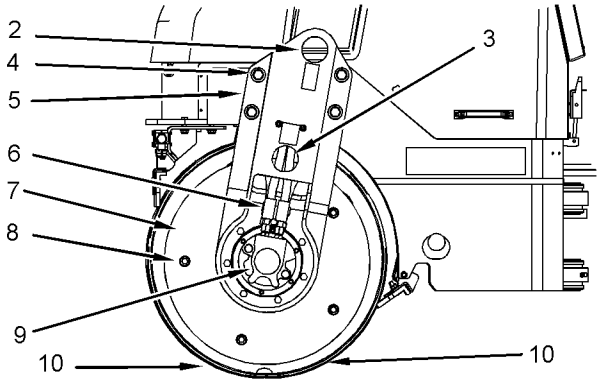


Illustration 81 g02079214

2. Block drums (10).

3. Secure support (5) with an appropriate strap or chain. Install the appropriate strap or chain through lifting eye (2) and the tie down location (3).

4. Remove four hoses (6) that are on hydraulic motor (9).

5. Remove four bolts (4) in order to remove support (5).

Note: Block the frame of the machine before removing support (5).

6. Remove five bolts (8) that secure plate (7) to the isolation mounts.

7. Remove plate (7) and remove hydraulic motor (9).

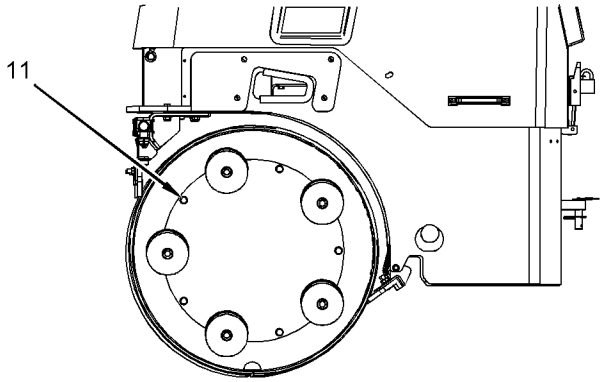


Illustration 82 g02079153

8. Remove five bolts on plate (11).

Note: Use Caterpillar 4C-9506 Retaining Compound when you reassemble bolts (11).

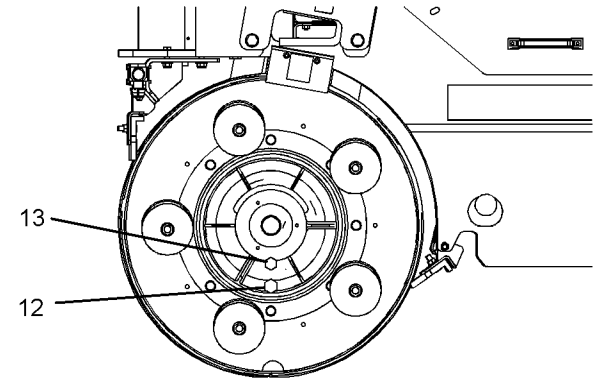


Illustration 83 g02079154

9. Remove drain plug (12) and allow the gear lubricant to drain.

10. Drain the oil in a suitable container.

11. Clean drain plug (12). Install drain plug (12).

12. Remove plug (13).

13. Add gear lubricant through plug (13).

Note: Refer to Operation and Maintenance Manual, “Capacities (Refill)” for the appropriate amount of gear lubricant.

14. Clean plug (13). Install plug (13).

15. Reverse the process in order to reassemble the drum.

i01953476

Engine Air Filter Primary Element - Clean/Replace

SMCS Code: 1054-510-PY; 1054-070-PY

NOTICE
Service the air cleaner only with the engine stopped. Engine damage could result.

Service the air cleaner filter element when the yellow piston on the engine air filter service indicator enters the red zone or the indicator reads 63.5 cm (25 inch) of water. Refer to Operation and Maintenance Manual, “Engine Air Filter Service Indicator - Inspect”.

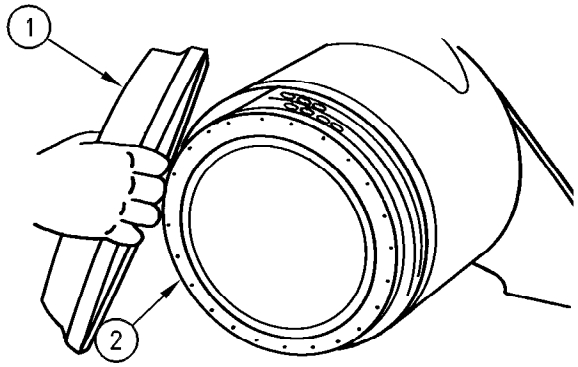


Illustration 84 g00102316

1. Remove cover (1) for the air filter housing.

2. Remove primary filter element (2) from the air filter housing.

3. Clean the inside of the air filter housing.

4. If the machine is equipped with a vacuator valve, clean the vacuator valve on the cover for the air filter housing.

5. Install a clean primary air filter element. Install the cover for the air filter housing.

Note: Refer to “Cleaning Primary Air Filter Elements”.

6. Reset the engine air filter service indicator.

7. Close the access door.

If the yellow piston in the indicator moves into the red zone after starting the engine or the exhaust smoke is still black after installation of a clean primary filter element, install a new primary filter element. If the piston remains in the red zone replace the secondary element.

Cleaning Primary Air Filter Elements

NOTICE
Caterpillar recommends certified air filter cleaning services available at participating Caterpillar dealers. The Caterpillar cleaning process uses proven procedures to assure consistent quality and sufficient filter life.

Observe the following guidelines if you attempt to clean the filter element:

Do not tap or strike the filter element in order to remove dust.

Do not wash the filter element.

Use low pressure compressed air in order to remove the dust from the filter element. Air pressure must not exceed 207 kPa (30 psi). Direct the air flow up the pleats and down the pleats from the inside of the filter element. Take extreme care in order to avoid damage to the pleats.

Do not use air filters with damaged pleats, gaskets, or seals. Dirt entering the engine will cause damage to engine components.

The primary air filter element can be used up to six times if the element is properly cleaned and if the element is properly inspected. When the primary air filter element is cleaned, check for rips or tears in the filter material. The primary air filter element should be replaced at least one time per year. This replacement should be performed regardless of the number of cleanings.

NOTICE
Do not clean the air filter elements by bumping or tapping. This could damage the seals. Do not use elements with damaged pleats, gaskets, or seals. Damaged elements will allow dirt to pass through. Engine damage could result.

Visually inspect the primary air filter elements before cleaning. Inspect the air filter elements for damage to the seal, the gaskets, and the outer cover. Discard any damaged air filter elements.

There are two common methods that are used to clean primary air filter elements:

- Pressurized air
- Vacuum cleaning

Pressurized Air

Pressurized air can be used to clean primary air filter elements that have not been cleaned more than two times. Pressurized air will not remove deposits of carbon and oil. Use filtered, dry air with a maximum pressure of 207 kPa (30 psi).

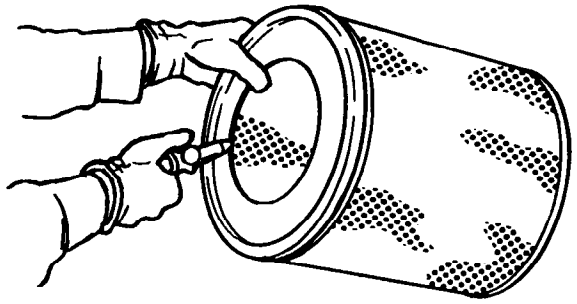


Illustration 85 g00281692

Note: When the primary air filter elements are cleaned, always begin with the clean side (inside) in order to force dirt particles toward the dirty side (outside).

Aim the hose so that the air flows inside the element along the length of the filter in order to help prevent damage to the paper pleats. Do not aim the stream of air directly at the primary air filter element. Dirt could be forced further into the pleats.

Vacuum Cleaning

Vacuum cleaning is another method for cleaning primary air filter elements which require daily cleaning because of a dry, dusty environment. Cleaning with pressurized air is recommended prior to vacuum cleaning. Vacuum cleaning will not remove deposits of carbon and oil.

Inspecting the Primary Air Filter Elements

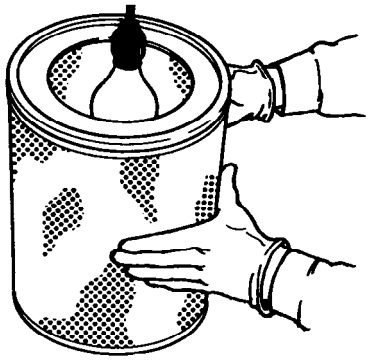


Illustration 86 g00281693

Inspect the clean, dry primary air filter element. Use a 60 watt blue light in a dark room or in a similar facility. Place the blue light in the primary air filter element. Rotate the primary air filter element. Inspect the primary air filter element for tears and/or holes. Inspect the primary air filter element for light that may show through the filter material. If it is necessary in order to confirm the result, compare the primary air filter element to a new primary air filter element that has the same part number.

Do not use a primary air filter element that has any tears and/or holes in the filter material. Do not use a primary air filter element with damaged pleats, gaskets or seals. Discard damaged primary air filter elements.

Storing Primary Air Filter Elements

If a primary air filter element that passes inspection will not be used, the primary air filter element can be stored for future use.

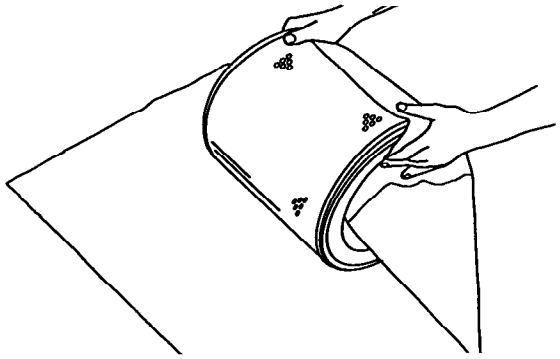


Illustration 87 g00281694

Do not use paint, a waterproof cover, or plastic as a protective covering for storage. An airflow restriction may result. To protect against dirt and damage, wrap the primary air filter elements in Volatile Corrosion Inhibited (VCI) paper.

Place the primary air filter element into a box for storage. For identification, mark the outside of the box and mark the primary air filter element. Include the following information:

- Date of cleaning
- Number of cleanings

Store the box in a dry location.

i01822684

Engine Air Filter Secondary Element - Replace

SMCS Code: 1054-510-SE

NOTICE
Always replace the secondary filter element. Never attempt to reuse the secondary filter element by cleaning the element.

When the primary filter element is replaced, the secondary filter element should be replaced.

The secondary filter element should also be replaced if the exhaust smoke is still black.

1. Open the access door on the front left side of the machine.
2. See Operation and Maintenance Manual, “Engine Air Filter Primary Element - Clean/Replace”. Remove the air cleaner cover from the air cleaner housing. Remove the primary filter element from the air cleaner housing.

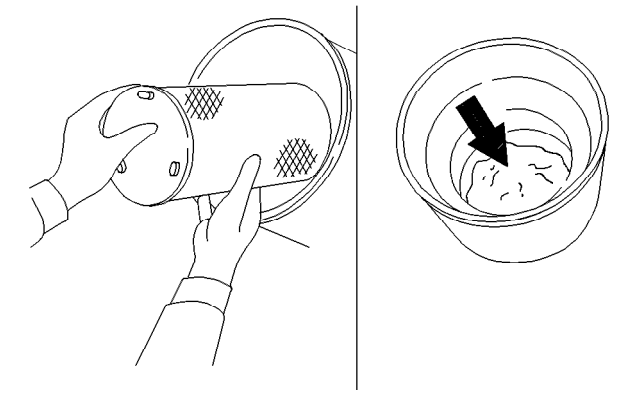


Illustration 88 g00101451

3. Remove the secondary filter element.

4. Cover the air inlet opening. Clean the inside of the air cleaner housing.

5. Remove the cover from the air inlet opening.

6. Install the new secondary filter element.

7. Install the primary filter element.

8. Install the air cleaner cover and close the latches securely.

9. Close the access door.

i01977910

Engine Air Filter Service Indicator - Inspect

SMCS Code: 7452-040

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

2. Start the engine.

3. Run the engine at high idle.

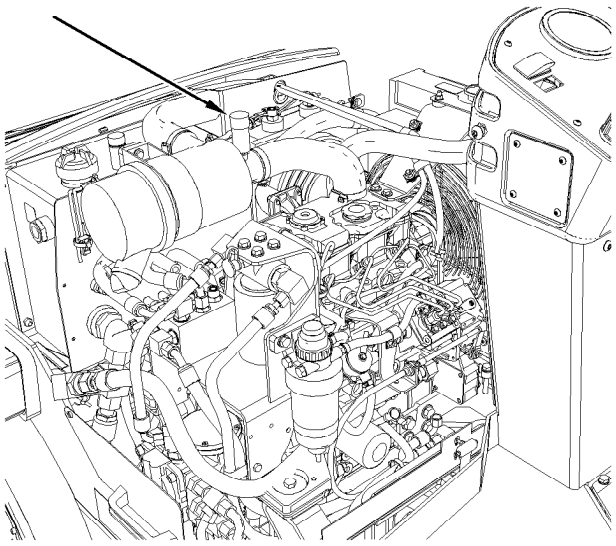


Illustration 89 g01033857

4. If the yellow piston in the engine air filter service indicator enters the red zone, service the air cleaner.

5. Stop the engine.

Note: See the Operation and Maintenance Manual, “Engine Air Filter Primary Element - Clean/Replace”. See the Operation and Maintenance Manual, “Engine Air Filter Secondary Element - Replace”.

6. Close the engine compartment.

i02166542

Engine Crankcase Breather - Clean/Replace

SMCS Code: 1317-510; 1317-070

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, “Caterpillar Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

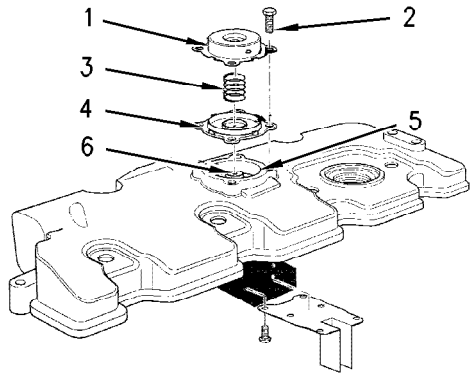


Illustration 90 g00827682

Typical example

(1) Breather cover
(2) Bolts for breather cover
(3) Spring
(4) Diaphragm and plate
(5) Cavity
(6) Vent hole

1. Loosen bolts (2) and remove breather cover (1) from the valve mechanism cover.

2. Remove spring (3). Remove the diaphragm and plate (4).

3. Clean vent hole (6). Clean cavity (5) for the breather assembly in the valve mechanism cover.

NOTICE
Make sure that the components of the breather assembly are installed correctly. Engine damage may occur if the breather assembly is not working properly.

4. Install a new diaphragm and plate (4) for the breather assembly. Install a new spring (3). Install breather assembly into cavity (5) of the valve mechanism cover.

5. Install breather cover (1) and four bolts (2). Tighten the bolts.

i01977919

Engine Cylinder Head Bolts - Check

SMCS Code: 1100-535-BC

This maintenance is recommended by Caterpillar as part of a lubrication and preventive maintenance schedule in order to help provide maximum engine life.

NOTICE
Only qualified service personnel should perform this maintenance. Refer to the Service Manual or your Caterpillar dealer for the complete valve lash adjustment procedure.

Operation of Caterpillarengines with improper valve adjustments can reduce engine efficiency. This reduced efficiency could result in excessive fuel usage and/or shortened engine component life.

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Engine Mounts - Inspect

SMCS Code: 1152-040

Open the engine compartment.

There are three engine mounts. There is one engine mount at the front of the engine. There are two engine mounts at the rear of the engine.

Engine vibration can be caused by improper mounting of the engine. Engine vibration can be caused by loose engine mounts or deteriorated engine mounts.

Inspect the engine mounts for deterioration.

Replace any engine mount that is deteriorated.

Inspect the engine mounts for correct bolt torque.

Tighten the mounts if the mounts are loose.

Close the engine compartment.

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Engine Oil Level - Check

SMCS Code: 1348-535-FLV

NOTICE
Do not under fill or overfill engine crankcase with oil. Either condition can cause engine damage.